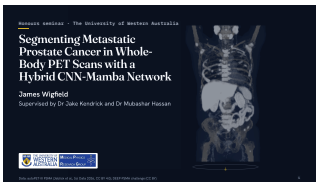
One Scan, Start to Finish: script

Segmenting Metastatic Prostate Cancer in Whole-Body PET Scans with a Hybrid CNN-Mamba Network

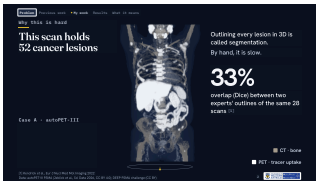
13 main slides · 1793 words · 13:38 spoken at 135 words a minute (including 21 s of clips without narration) · target 13:52



`Slide 1 · 20 s · ends at 0:20 · 36 words`

1. On entry: the case A scan turns slowly behind the title. Introduce yourself and the supervisors.

Hi everyone, I'm James. My honours project asks whether a newer kind of neural network, called Mamba, can help find prostate cancer lesions in whole-body PET scans. I was supervised by Jake Kendrick and Mubashar Hassan.

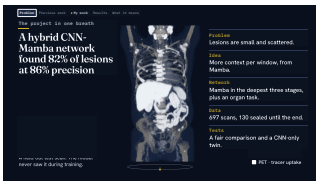


`Slide 2 · 74 s · ends at 1:34 · 158 words`

This scan holds 52 cancer lesions

1. On entry: the scan clip plays (fades in, one turn). Narrate over it, then ask the room to find the lesions.
2. Click: segmentation, outlining by hand is slow.
3. Click: the inter-observer Dice from the 2022 study.

`CLIP PLAYS, 8.8 S, KEEP TALKING` This is a whole-body PSMA PET/CT scan from my held-out test set. The model never saw it, and I picked it by a rule before looking. PSMA is a protein found on prostate cancer cells. A radioactive tracer binds to it, PET shows where the tracer collects, and CT shows the anatomy. Bright means tracer uptake. Much of that is normal: salivary glands, liver, kidneys, gut and bladder. The cancer is the scattered spots, mostly in the bones. Each spot is called a lesion, and this scan has 52 of them. Can you find them? `PAUSE` `CLICK` To measure the disease, someone has to outline every lesion in 3D. That outlining is called segmentation, and by hand it's slow. `CLICK` Experts also disagree. In a 2022 study, two experts outlined the same 28 scans. Their outlines overlapped by only 33%. That overlap score is called Dice. So a tool has to find the lesions and outline them well enough to measure. `(cut if behind)`

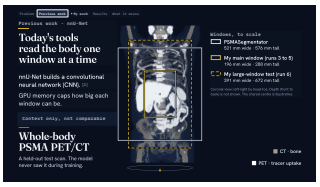


`Slide 3 · 36 s · ends at 2:10 · 80 words`

A hybrid CNN-Mamba network found 82% of lesions at 86% precision

1. On entry: the five overview lines build in beside the scan.

Here's the whole project in one breath. I built a hybrid network that adds Mamba to its deepest stages. I used 697 public scans and locked 130 of them away until the very end. On those it found 82% of the lesions, its sensitivity, and 86% of what it flagged was real, its precision. The gold dot in the bar at the top marks my own work. `(cut if behind)` From here, I'll follow this one scan through the model, then zoom out.

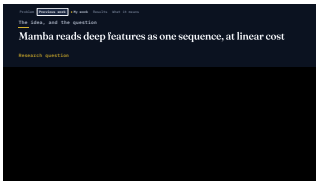


`Slide 4 · 70 s · ends at 3:20 · 155 words`

Today's tools read the body one window at a time

1. On entry: nnU-Net reads one window at a time.
2. Click: PSMASegmentator's window draws to scale.
3. Click: the 2022 cascade card (context only).
4. Click: this project's two windows in gold (small: runs 3 to 5; large: run 6).

The standard tool for this job is nnU-Net. It builds a convolutional neural network, or CNN, which looks at the image through small local filters. A whole-body scan won't fit in GPU memory, the graphics card's memory, so the network reads it one 3D window at a time. `CLICK` This box is PSMASegmentator's window, drawn to scale on our scan. PSMASegmentator is the released nnU-Net model for PSMA PET, and it's what I compare against. It can't use anything outside the window it's reading. `(cut if behind)` `CLICK` Before it came the 2022 nnU-Net cascade by Kendrick et al. On its own test set it found 73.0% of lesions, and 88.2% of what it flagged was real. Those figures are context only, because its test set and scoring rule aren't mine. `CLICK` The gold boxes are my own windows. My models are numbered as training runs: the final one, run 5, averages five models, and run 6 tests this larger window.



`Slide 5 · 78 s · ends at 4:38 · 163 words`

Mamba reads deep features as one sequence, at linear cost

1. On entry: Mamba against attention, schematic. Point at the web.
2. Click: the U-Mamba warning card.
3. Click: the research question in large type. Pause and let them read.
4. Click: H1 to H4, all pending.

The idea is to let the network see more inside each window, using Mamba. Mamba is a sequence model. It reads a long line of tokens in order and carries a small memory along. Here a token is one cell of a small 3D grid of features inside the network. Its cost grows linearly with length: double the sequence and the work doubles. `POINT AT THE WEB` Attention, the usual alternative, compares every token with every other, so doubling the length quadruples the work. `CLICK` But U-Mamba, an earlier hybrid, was retested with its Mamba layers switched off. The Mamba layers made no difference. So any claim for Mamba needs a twin without it. `CLICK` Hence my research question. `PAUSE` In short: does it beat PSMASegmentator, and is Mamba the reason? `CLICK` I split it into four predictions. H1 is detection, H2 the training loss, and H3 speed and memory. H4 asks whether a bigger window helps the hybrid more than its twin. Each one gets a verdict near the end. `(cut if behind)`

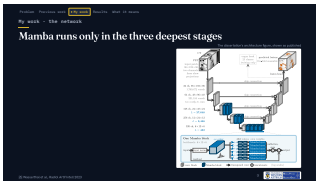


`Slide 6 · 52 s · ends at 5:30 · 114 words`

697 scans, 130 locked away until the very end

1. On entry: every scan as a tile, autoPET-III filled, DEEP-PSMA hollow.
2. Click: the sealed test tiles move into the vault.
3. Click: the pool regroups into five folds.
4. Click: the pre-registration card.

For data, I combined two public datasets, 697 scans in all. autoPET-III has 597 scans from Munich. Their lesions were outlined by a threshold, then corrected by an expert. DEEP-PSMA has 100 patients with advanced disease, labelled by a threshold on tracer uptake. `CLICK` First I sealed 130 scans away. Nothing trained or was tuned on them. `CLICK` The other 567 went into five-fold cross-validation. Each model trains on four folds and is checked on the fifth, so every pool scan is checked once. `CLICK` Before opening the seal, I wrote down the probability cut-off, the smallest blob I'd keep, and the numbers I'd report. That's what pre-registered means here. Then the test set was scored once. `(cut if behind)`

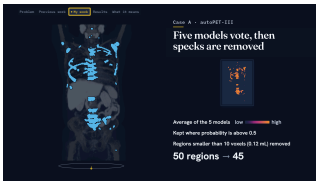


`Slide 7 · 64 s · ends at 6:34 · 142 words`

Mamba runs only in the three deepest stages

1. On entry: option F, input and the convolutional stages.
2. Click: the three Mamba stages and their sequence lengths.
3. Click: one Mamba block, six directions.
4. Click: decoder and heads; the organ head is training only.
5. Click: the whole figure.

Here's the network, from the dissertation. It's U-shaped: the window goes down through five stages and back up to a lesion map. `POINT AT THE INPUT` Two channels go in, CT and PET. The first two stages are plain convolutions. There the sequence would be far too long, and fine texture suits a convolution anyway. `CLICK` Mamba blocks sit only in the three deepest stages, where the grid is small. At the bottom the sequence is just 432 tokens. `CLICK` Inside a block, the grid is flattened and read in six directions by separate Mamba layers. Their outputs are summed and added back. `CLICK` On the way up, skip connections bring back the fine detail. A second output learns anatomy, but only during training. When I added it, the spots it wrongly flagged on seven validation scans with no lesions fell from 21 to 7. `CLICK` That's the whole network. `(cut if behind)`

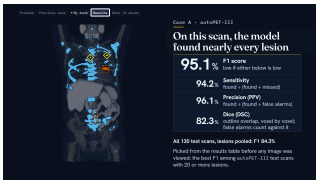


`Slide 8 · 44 s · ends at 7:18 · 94 words`

Five models vote, then specks are removed

1. On entry: the windows clip. Keep talking.
2. Click: the five-models clip; the fold maps fan out.
3. Click: the threshold clip; average, threshold, specks removed. Pause for the specks.

`CLIP PLAYS, 5.5 S, KEEP TALKING` Back to our scan. It goes through the network one window at a time: 160 windows here, each overlapping its neighbours by half. The glow is one model's running estimate, filling in. `CLICK` `CLIP PLAYS, 6.6 S, KEEP TALKING` I trained five of these models, one per fold. Each gives its own probability map. Averaging models like this is called an ensemble. `CLICK` `CLIP PLAYS, 5.8 S` Voxels, the 3D pixels, with an average above 0.5 are kept. Then specks under 10 voxels are dropped. That size is about the limit of what PET can resolve, and it scored best in validation. `PAUSE` 50 regions become 45. `(cut if behind)`

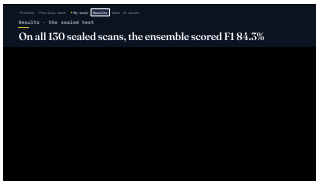


`Slide 9 · 56 s · ends at 8:14 · 124 words`

On this scan, the model found nearly every lesion

1. On entry: the prediction clip turns.
2. Click: the ground-truth clip, with verdicts. Name the three shapes.
3. Click: the score-card clip. Define the metrics on it.

`CLIP PLAYS, 9.6 S, KEEP TALKING` This is the final prediction: every region the model marks as a lesion. `CLICK` `CLIP PLAYS, 8.8 S, KEEP TALKING` Now the expert-corrected outlines go on. A circle is a lesion found, which means some predicted voxel touches it. A dashed square is a lesion missed. A diamond is a false alarm: a predicted region that touches no lesion at all. Two of the three misses are specks smaller than the size filter. `(cut if behind)` `CLICK` `CLIP PLAYS, 5.2 S, KEEP TALKING` Now the scores. Sensitivity is the share of lesions found, and precision the share of finds that were real. F1 combines the two, so it's low if either one is low. Dice is the outline overlap from earlier. The rule picked the best F1 among autoPET scans with 20 or more lesions, so this is the good end.

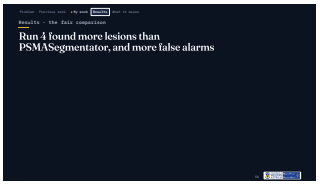


`Slide 10 · 50 s · ends at 9:04 · 114 words`

On all 130 sealed scans, the ensemble scored F1 84.3%

1. On entry: the sealed scans, case A marked.
2. Click: the pooled lesion bars (found, missed, false alarms).
3. Click: pooled and in-domain, with the interval.

Now zoom out to all 130 sealed scans. This is the main result, scored once under the protocol I fixed in advance. The ensemble's F1 was 84.3%. `CLICK` Across those scans there are 4,554 lesions. It found 3,754 and missed 800, and it raised 601 false alarms. That's 82.4% sensitivity and 86.2% precision. `CLICK` Two words matter here. Pooled means the lesions are added up across scans, so the busiest scans count most. The mean of the per-scan F1 is lower, about 76.8%. In-domain means the test scans come from the same two datasets as training. The 95% interval on F1 runs from 81.4 to 86.5. Both datasets land on the same F1, by different routes. `(cut if behind)`



`Slide 11 · 82 s · ends at 10:26 · 182 words`

Run 4 found more lesions than PSMASegmentator, and more false alarms

1. On entry: the underdog card.
2. Click: the pooled result and the trade.
3. Click: one dot per scan.

Now the fair comparison with PSMASegmentator. It trained on every autoPET scan in my test set, so I can't score it there. Instead I used the 100 DEEP-PSMA scans, which neither model trained on. My side is run 4. It isn't my final model, just an earlier single hybrid trained on autoPET alone. It's the underdog: PSMASegmentator has more training scans, five models instead of one, eight mirrored passes and a bigger network. Mirrored passes, or test-time augmentation, run each window again flipped and average the results. So a win for run 4 is the safer finding. `CLICK` Even so, run 4's pooled F1 was 3.0 points higher, and the interval excludes zero. It found more lesions, with sensitivity up 8.1 points. It also raised more false alarms, with precision down 6.4. So H1 holds only as a trade. `CLICK` Scan by scan, there's no clear difference. Run 4 was better on 48 and worse on 50. It lost on most scans with few lesions and won on most with many. `(cut if behind)` Pooled F1 weights the busy scans, so that's where the gain comes from.

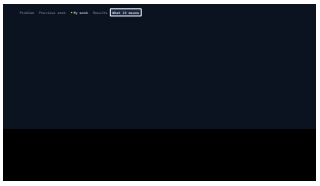


`Slide 12 · 98 s · ends at 12:04 · 217 words`

On unseen scans, a CNN-only twin did as well

1. On entry: the twin; small-window scores only.
2. Click: the larger window, both lines drop.
3. Click: the efficiency line and the abstract timing note.
4. Click: the scoreboard replaces the chart, H1 to H4 resolved.

Now the test of Mamba itself. I trained a twin: the same network and recipe with every Mamba block removed. That's called an ablation. At the small window, on the same 100 unseen scans, the twin scored 82.0 against the hybrid's 81.4. Both sit above PSMASegmentator, though only the hybrid's gap was tested with an interval. On the autoPET test scans the hybrid was +1.79 points ahead, which passes my rule but is small. So the Mamba blocks added little, and the gain isn't credited to them. `CLICK` Then I gave both networks the larger window. If Mamba makes use of context, the hybrid should gain more. `(cut if behind)` But the larger window cost the hybrid 4.55 points, and the twin only 0.97. That difference was pre-registered, and it counts as real. One possible reason is that a much longer sequence asks too much of Mamba's small memory. `CLICK` At equal passes, the hybrid was no faster in mean time and no lower in peak memory. The abstract's 32 against 63 seconds, and 18.1 s against 43.8 s in the dissertation, both come from PSMASegmentator's eight-pass test-time augmentation. `CLICK` So the scoreboard. H1 holds, but only as a trade. H2, tested only partly, and H3 weren't supported. H4 came out reversed on the primary set: the bigger window hurt the hybrid more.

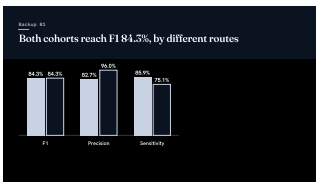


`Slide 13 · 108 s · ends at 13:52 · 214 words`

Some scans still fail, and the next test is SCGH data

1. On entry: case B against its ground truth.
2. Click: the limitations.
3. Click: what it means, the contribution and the next step.
4. Click: the recorded run in the app. Name its colours, then let it play.
5. Click: case A beside case B with the conclusion; thank you and questions.

`CLIP PLAYS, 6.2 S, KEEP TALKING` Here's the other end. Case B, picked by rule, has the lowest F1 among scans where any lesion was found. Both of its lesions were found, but the model raised 14 false alarms, so its precision was only 12.5%. It isn't the worst scan: on nine scans the model found none of the lesions. `CLICK` So, the limits. I tested on two public datasets, never on the Sir Charles Gairdner Hospital cohort my research proposal named. Each run 6 cell is a single training run. And only 8 test scans have no lesions. `CLICK` What it means: on unseen scans, run 4, my single hybrid, found more lesions than PSMASegmentator, with more false alarms. There the CNN-only twin did as well, so I don't credit the gain to Mamba. The project offers a fair test on scans unseen by both, and a pre-registered test of what the Mamba blocks add. The next step is to retrain and score it on that hospital cohort. Once the code is released, the UWA Medical Physics Research Group can do that. `CLICK` `CLIP PLAYS, 18 S` Here's the model running in my app, sped up 6x. The app's colours: green found, cyan missed, magenta false alarm. `LET IT PLAY` `CLICK` That was case A and case B, from the same model. `(cut if behind)` Thank you, and I'm happy to take questions.



`Backup B1 · 76 words`

Both cohorts reach F1 84.3%, by different routes

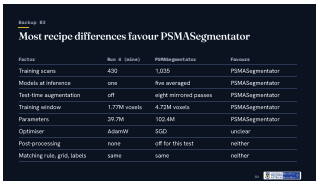
The two cohorts label lesions differently, so I report them separately. autoPET-III outlines were corrected slice by slice by an expert. DEEP-PSMA marks every voxel at SUV 3 or more, with normal uptake painted out, so a scan splits into about 68 small pieces. The likely reason for the gap: on DEEP-PSMA nearly everything the model predicts lands on disease, so precision is 96.0%, but it misses many small pieces. Both land on the same F1.



`Backup B2 · 63 words`

Cross-validation agreed with the sealed test

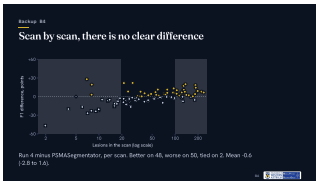
Before the test, each of the 567 pool scans was predicted by the fold model that never trained on it. That pooled F1 was 81.8%, with a fold-to-fold spread of 1.5 points, and it sits inside the test interval of 81.4 to 86.5. A single fold model scored 83.1 on average on the test set, so averaging the five added about 1.2 points.



Backup B3 · 70 words

Most recipe differences favour PSMASegmentator

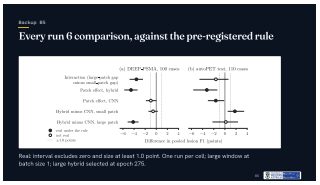
This is the honest list of what differs. PSMASegmentator trained on 1,035 scans against my 430, averages five models with eight mirrored passes, uses a bigger window, and has 102.4M parameters against 39.7M. It also has deep supervision and a misalignment augmentation that I did not port. Most factors favour it, so my gain is more likely to understate than overstate the recipe, but none was tested on run 4.



Backup B4 · 65 words

Scan by scan, there is no clear difference

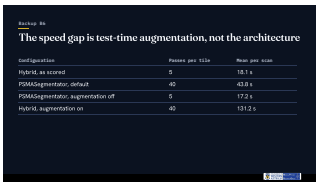
Each dot is one of the 100 DEEP-PSMA scans. Pooled F1 weighs lesion-heavy scans most, so the pooled gain comes from the right of this plot. Run 4 was below PSMASegmentator on 18 of the 22 scans with under 20 lesions and above it on 23 of the 28 with 100 or more. The mean per-scan difference was -0.6 points, with an interval spanning zero.



Backup B5 · 62 words

Every run 6 comparison, against the pre-registered rule

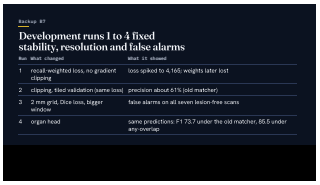
These are all five comparisons I committed to before scoring, from the dissertation. On DEEP-PSMA the interaction was -3.57 points, from -4.72 to -2.48, a real effect under the rule. On autoPET it was -1.62, no clear effect. The caveats: one training run per cell, the large window trained at batch size 1, and the large hybrid was selected at epoch 275.



Backup B6 · 73 words

The speed gap is test-time augmentation, not the architecture

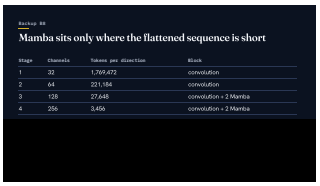
Timed on one RTX 4090 over 20 test scans, each model already loaded. As scored, the hybrid took 18.1 s against 43.8 s for PSMASegmentator, but that gap is its eight-pass augmentation. At equal passes the hybrid was no faster in mean time and used 5.2 GiB of memory against 4.5 GiB. The abstract quoted 32 against 63 seconds from an earlier run on my home RTX 5070 Ti; the same caveat applies.



Backup B7 · 76 words

Development runs 1 to 4 fixed stability, resolution and false alarms

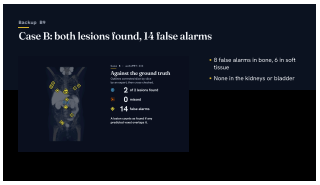
Run 1 had no gradient clipping and lost about three of its twelve hours to loss spikes. Run 2 added clipping, but the recall-weighted loss left precision at about 61%. Run 3 moved to the finer grid and Dice, then flagged every lesion-free validation scan. Run 4 added the organ head. Runs 2 to 4 were scored under an older one-to-one matcher; on identical run 4 predictions it gave 73.7 where the adopted rule gives 85.5.



Backup B8 · 64 words

Mamba sits only where the flattened sequence is short

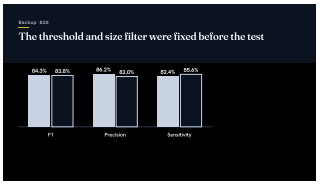
Each deeper stage halves the grid, so the sequence a Mamba layer would scan shrinks fast. The shallow stages would be at least 221,184 tokens per direction, and their features are fine texture a convolution handles cheaply. In the deep three, each block scans six directions and sums them. I never tested the alternatives, such as alternating directions across layers or a larger state.



Backup B9 · 68 words

Case B: both lesions found, 14 false alarms

Case B was chosen by rule before I looked: the lowest F1 among scans where F1 is defined. It is not the worst scan, because nine scans had no lesion found and so have no F1. Both of its lesions were found. Its 14 false alarms were 8 in bone and 6 in soft tissue, and none sat in the kidneys or bladder, which the organ task targets.



Backup B10 · 64 words

The threshold and size filter were fixed before the test

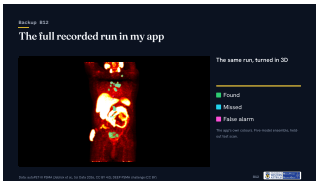
The operating point came from the five-fold validation sweep, before any test inference. With the 10-voxel filter, F1 moved by less than a point across thresholds, so I kept 0.5. The protocol fixed two rows: the filtered headline and an unfiltered row, because PSMASegmentator has no filter. On the test set the filter trades a little sensitivity for precision, and both rows are reported.



Backup B11 · 27 words

References

These are the works cited on the slides, in the order they appear, in the same IEEE style as the dissertation. The dissertation has the full list.



Backup B12 · 62 words

The full recorded run in my app

This is a real run of the five-model ensemble in the app I built, on a held-out test scan, sped up 6x. It needs a CUDA GPU, so I recorded it on my PC. The colours are the app's own: found, missed and false alarm. The code will be released so the UWA Medical Physics Research Group can retrain and score it.